

Predictive Method Selection Module

Module: PMSM — Predictive Method Selection Module

Parent Standard: PMES — Prediction Methodology Standard

Architecture: PGA™ — Prediction Governance Architecture

Category: Governance & Enforcement

Subcategory: Prediction Governance

Governed Space: Predictive Method Selection

Primary Governed Object: Selected Predictive Method

Module Type: Core Module

Canonical Language: UCL — Universal Canonical Language

Protection: MIP® — Methodological Intellectual Property

1. Canonical Definition

The Predictive Method Selection Module (PMSM) governs the identification, comparison, selection, justification, and selection-state governance of the predictive method or combination of methods intended to serve a governed predictive frame.

PMSM establishes whether a predictive method has been deliberately selected in relation to the Prediction Object, Predictive Purpose, Prediction Target, Prediction Horizon, Predictive Boundary, and relevant methodological requirements rather than adopted merely because it is available, familiar, popular, institutionally preferred, historically successful, or technically capable of producing an output.

Method Availability ≠ Method Applicability

2. Operational Role

PMSM is the first Core Module within PMES.

It receives the governed predictive frame established by POPS and determines which predictive method or candidate combination of methods will proceed into methodological structuring.

Its principal transition is:

Governed Predictive Frame → Candidate Predictive Methods → Selected Predictive Method

PMSM governs method selection.

It does not establish the complete methodological structure or determine whether the final methodology is sufficient.

3. Internal Module Flow

→ Predictive Method Selection → Methodological Structure Definition → Predictive Logic Documentation
→ Method Applicability → Methodology Version Governance

Current Position: Predictive Method Selection



The selected method proceeds into:

4. Operational Space

PMSM operates within the Predictive Method Selection space.

It governs:

method requirement identification; candidate method identification; candidate method differentiation; method comparison; method-selection criteria; predictive-frame alignment; method capability relevance; methodological limitations at selection; dependency awareness; multi-method selection; selection justification; selection state.

PMSM does not determine which method is universally best.

It governs which method is selected for the specific predictive context.

5. Operational Scope

PMSM may govern selection involving:

statistical methods; mathematical forecasting; econometric methods; physical models; machine-learning models; AI systems; human forecasting; expert judgment; prediction markets; forecasting tournaments; ensembles; simulation-supported methods; multi-agent prediction; hybrid human-machine methodologies; domain-specific forecasting methods; other sufficiently identifiable predictive mechanisms.

PGA™ remains method-neutral.

No method receives governed status merely because of its technological class.

6. Method Requirement Identification

Before selecting a method, PMSM must establish the predictive requirements the method is expected to serve.

These requirements originate from the governed predictive frame, including:

Prediction Object; Predictive Purpose; Prediction Target; Prediction Horizon; Predictive Boundary.

Selection must begin from the predictive problem rather than from the preferred tool.

Predictive Problem → Method Selection

not:

Available Method → Redefined Predictive Problem

7. Candidate Method Identification

PMSM must identify the method or methods materially considered for predictive use.

Where meaningful alternatives exist, the selection process should preserve sufficient differentiation between candidate approaches.

A candidate method does not become selected merely because it is capable of generating an output.

Candidate Method ≠ Selected Predictive Method

8. Method Differentiation

Candidate methods must remain distinguishable where their predictive mechanisms, dependencies, assumptions, limitations, outputs, or applicability materially differ.

For example, an expert forecast, statistical model, AI model, prediction market, and simulation-supported forecast may address the same Prediction Object while representing fundamentally different predictive methods.

They must not be treated as methodologically equivalent solely because they produce predictions concerning the same target.

Same Prediction Target $\neq$ Same Predictive Method

9. Selection Criteria

PMSM requires the material criteria governing method selection to remain identifiable.

Criteria may concern:

suitability to the Prediction Object; alignment with Predictive Purpose; target characteristics; Prediction Horizon; predictive boundary; required resolution; available information; methodological limitations; reproducibility requirements; interpretability requirements; operational constraints; dependency requirements; other contextually material factors.

Selection criteria must relate to the predictive problem.

10. Predictive-Frame Alignment

The selected method must be capable of addressing the predictive frame established by POPS.

A method suitable for one target, horizon, population, environment, or predictive purpose may be inappropriate for another.

PMSM therefore requires explicit consideration of method-to-frame alignment.

General Predictive Capability $\neq$ Contextual Methodological Suitability

11. Method Capability

A method must possess capabilities relevant to the predictive task.

However, capability alone does not establish selection legitimacy.

A large AI model may be capable of producing a prediction.

A prediction market may be capable of producing a probability-like price.

An expert may be capable of expressing a forecast.

A statistical model may be capable of extrapolating historical patterns.

None of these facts independently establish that the method is suitable for the governed prediction.

Ability to Produce an Output $\neq$ Reason to Select the Method

12. Historical Performance

Historical predictive performance may inform method selection where relevant.

It must not become the sole automatic justification for selecting a method without considering whether historical predictive conditions remain applicable.

A method may perform well in one environment and poorly in another.

Historical Performance $\neq$ Future Applicability

Past Accuracy $\neq$ Universal Methodological Suitability

Performance governance itself belongs to POFPS — Prediction Outcome & Performance Standard .

13. Method Popularity and Institutional Preference

A method must not be selected solely because it is:

widely used; institutionally preferred; commercially dominant; technologically fashionable; provided by a major organization; considered industry standard; computationally advanced.

Such characteristics may form part of contextual assessment but cannot substitute for methodological selection logic.

Method Popularity $\neq$ Method Suitability

Institutional Reputation $\neq$ Predictive Applicability

14. Human Method Selection

Human forecasting and expert judgment may constitute legitimate predictive methods.

PMSM must not reject human methods solely because they contain judgment.

Where human judgment is selected, the reason for using it within the predictive context must remain governable.

Likewise, expertise alone does not automatically establish suitability.

Expert Status $\neq$ Methodological Suitability

15. AI Method Selection

AI systems may constitute or support predictive methods.

Selection of an AI system must not be justified solely by model scale, benchmark performance, technological novelty, provider reputation, or general intelligence capability.

The relevant governance question remains whether the AI-based method is appropriate for the specific predictive frame.

Model Capability $\neq$ Prediction-Specific Suitability

AI behavioral governance remains outside PMSM.

16. Prediction Market Method Selection

Prediction markets may serve as predictive mechanisms where market-based aggregation is relevant to the predictive purpose.

Their selection must not assume that market price automatically represents a governed probability estimate, population belief, independent collective judgment, or methodologically superior prediction.

PMSM governs whether a market-based mechanism is an appropriate selected method.

Subsequent PGA™ standards govern its basis, collective composition, probability representation, and performance.

Market Price Availability $\neq$ Methodological Suitability

17. Simulation-Supported Method Selection

Simulation may support a predictive methodology where simulated pathways, states, scenarios, or distributions are relevant.

Selection of simulation does not convert simulation into prediction.

Simulation Capability $\neq$ Prediction Capability

Simulation Output $\neq$ Prediction

SIMULOS® retains governance ownership of simulation.

PMSM governs only the decision to use simulation-derived capability within Prediction Methodology.

18. Multi-Method Selection

A prediction may legitimately require multiple methods.

PMSM must establish whether the selected methods are intended to operate:

independently; sequentially; comparatively; as an ensemble; through aggregation; through human-machine combination; through another governed multi-method relationship.

Detailed methodological structure belongs to MSDM.

PMSM establishes that multiple methods have been deliberately selected.

Multiple Selected Methods $\neq$ Governed Combined Methodology

19. Method Selection Dependencies

Selection may depend upon access to:

particular information; data; models; software; infrastructure; expertise; markets; external services; computational resources; simulation systems; other methodological resources.

Material dependencies known at selection must remain identifiable where their absence could materially affect whether the selected method remains usable or suitable.

20. Method Selection Constraints

Predictive environments may impose constraints including:

time; computational capacity; available information; cost; operational accessibility; model availability; expertise; regulatory restrictions; technical limitations.

PMSM may recognize these constraints without treating the most convenient available method as automatically suitable.

Operational Convenience $\neq$ Methodological Suitability

21. Selection Trade-Offs

No predictive method is required to dominate every relevant methodological dimension.

Selection may involve trade-offs between characteristics such as:

accuracy; interpretability; speed; cost; resolution; scalability; reproducibility; adaptability; data dependency; computational requirements.

Material trade-offs affecting method selection must remain identifiable.

PMSM does not establish one universal hierarchy among them.

22. Method Selection Justification

A selected method must have a sufficiently identifiable methodological reason for selection.

The justification should establish why the method is appropriate for the governed predictive frame rather than merely describing the method itself.

Method Description $\neq$ Method Selection Justification

A technically detailed method can still lack a legitimate reason for being selected for a particular prediction.

23. Selection Reconsideration

Selection may require reconsideration where material predictive conditions change before methodology formation or use.

Potential triggers may include:

Prediction Object change; Predictive Purpose change; Prediction Target change; Prediction Horizon change; Predictive Boundary change; method unavailability; newly identified limitation; dependency failure; emergence of materially more appropriate methods.

PMSM governs method-selection reconsideration.

Lifecycle prediction revision remains governed elsewhere.

24. Selection Sufficiency

Predictive method selection is sufficiently governed when:

predictive requirements are identifiable; candidate method is identifiable; material alternatives are distinguishable where relevant; selection criteria are identifiable; selected method aligns with the governed predictive frame; material known limitations are considered; material dependencies and constraints are recognized; multi-method selection is explicit where applicable; selection justification is preserved.

Selection sufficiency does not establish complete Method Applicability.

That later determination belongs to MAPM.

25. Governance Requirements

PMSM requires that:

1. method selection begin from the governed predictive frame; 2. the selected method be explicitly identifiable; 3. material candidate methods remain distinguishable where relevant; 4. selection criteria relate to the predictive problem; 5. method capability not substitute for method suitability; 6. historical performance not establish universal applicability; 7. popularity or institutional reputation not substitute for selection logic;

8. material dependencies and constraints be recognized; 9. multi-method selection remain explicit; 10. material trade-offs remain identifiable; 11. selection justification be preserved; 12. material changes capable of invalidating the selection trigger reconsideration.

26. Governance Logic

PMSM applies the following operational logic:

Governed Predictive Frame

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Predictive Requirements Identified

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Candidate Method(s) Identified

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Candidate Differentiation

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Selection Criteria Established

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Predictive-Frame Alignment Assessment

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Dependencies / Constraints / Limitations Considered

↓

Trade-Off Assessment where applicable

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Method Selection Justification

↓

Selected Predictive Method

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MSDM — Methodological Structure Definition Module

Where selection cannot be sufficiently justified:

Selection Governance Gap → Reconsideration / Alternative Method Assessment

27. Minimum Implementation Framework

A minimum implementation of PMSM must establish:

A. Predictive Frame Reference

Reference to the governed POPS output.

B. Predictive Requirements

Material requirements the method is expected to satisfy.

C. Candidate Method

The predictive method proposed for selection.

D. Alternative Methods

Material alternatives considered where relevant.

E. Selection Criteria

The criteria governing selection.

F. Frame Alignment

Relationship between the candidate method and Prediction Object, Purpose, Target, Horizon, and Boundary.

G. Known Method Limitations

Material limitations identifiable at selection.

H. Dependencies and Constraints

Material conditions affecting method selection.

I. Selection Trade-Offs

Material trade-offs where applicable.

J. Selection Justification

The governed reason for selecting the method.

K. Selection State

At minimum:

Unselected → Candidate → Under Assessment → Selected → Reconsideration Required → Replaced

Only a sufficiently Selected method proceeds into methodological structure definition.

28. Governance Outputs

PMSM may produce:

Selected Predictive Method; candidate-method record; selection criteria; method-selection justification; predictive-frame alignment reference; known limitation context; dependency context; constraint context; trade-off context; multi-method selection state;

selection state; unresolved selection requirements.

The primary governed output is:

Selected Predictive Method

This becomes the principal input into:

MSDM — Methodological Structure Definition Module

29. Critical Distinctions

Predictive Problem ≠ Available Method

Candidate Method ≠ Selected Predictive Method

Same Prediction Target ≠ Same Predictive Method

Method Availability ≠ Method Applicability

General Predictive Capability ≠ Contextual Methodological Suitability

Ability to Produce an Output ≠ Reason to Select the Method

Historical Performance ≠ Future Applicability

Past Accuracy ≠ Universal Methodological Suitability

Method Popularity ≠ Method Suitability

Institutional Reputation ≠ Predictive Applicability

Expert Status ≠ Methodological Suitability

Model Capability ≠ Prediction-Specific Suitability

Market Price Availability ≠ Methodological Suitability

Simulation Output ≠ Prediction

Multiple Selected Methods ≠ Governed Combined Methodology

Operational Convenience ≠ Methodological Suitability

Method Description ≠ Method Selection Justification

30. Enterprise Application

PMSM creates a governed decision layer between a predictive problem and the technology, model, expert, market, analytical mechanism, or combination of methods selected to address it.

This is particularly important in environments where organizations possess many predictive capabilities and may otherwise select methods because they are already licensed, operationally convenient, technologically fashionable, provided by a preferred vendor, or historically successful.

PMSM reverses that logic:

Define the predictive problem first. Select the method second.

31. Use Case 1 — Selecting AI Because It Is the Most Powerful Available Model

An organization must predict a high-impact operational condition and automatically selects its most advanced AI model because it performs strongly across general benchmarks.

The model can generate a convincing forecast and numerical probability.

However, the organization has not established whether the model is suitable for the specific Prediction Target, whether its predictive capabilities apply across the required horizon, whether the relevant operational context falls within its effective predictive conditions, or whether another method would be more appropriate.

The governance question is:

Should the most capable available model be selected when its capability has not been shown to correspond to the predictive problem being governed?

PMSM requires selection to be justified against the governed predictive frame.

Model Capability $\neq$ Prediction-Specific Suitability

32. Use Case 2 — Choosing Between Expert Forecasting, AI, and a Prediction Market

An organization needs a prediction concerning a consequential future event.

Three methods are available:

an expert group with direct domain knowledge,

an AI forecasting system processing large volumes of information,

and a prediction market producing a continuously changing market-implied probability.

All three generate plausible predictive outputs.

Their methodologies, information access, dependencies, incentives, limitations, and relationship to Operational Reality are materially different.

The governance question is:

On what methodological basis should one predictive method be selected—or multiple methods combined—when each produces a prediction but represents a fundamentally different predictive mechanism?

PMSM requires the selection criteria, predictive-frame alignment, material trade-offs, dependencies, and selection justification to be established before one method is treated as the governed method.

Multiple Available Predictions $\neq$ Methodological Equivalence

33. Canonical Boundary

PMSM governs the identification, differentiation, comparison, selection, justification, dependencies, constraints, trade-offs, and selection state of predictive methods intended to serve a governed predictive frame. It does not govern detailed methodological structure, complete predictive logic documentation, final Method Applicability determination, methodology version governance, Prediction Basis, assumptions, prediction formation, probability, feasibility, collective prediction legitimacy, or predictive performance.

PMSM governs which predictive method is selected and why .

It does not establish what that method will ultimately predict .

Canonical Closing Statement

The strongest predictive tool is not necessarily the right predictive method.

PMSM requires the method to be selected because it fits the prediction — not the prediction to be reshaped because a method is available.

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